Quantifying Differences in Hydrogen Peroxide and L-lactate Production across Oral Streptococci using an Optimized Reflectometric Assay

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Abstract

Interactions among bacteria in the oral microbiome influence the development and progression of periodontal disease. *Streptococcus*, the most abundant bacterial genus in the oral cavity, interacts with *Aggregatibacter actinomycetemcomitans* (*Aa*), a pathogen associated with a severe form of periodontitis. For example, hydrogen peroxide (H_2O_2) production from the symbiont *Streptococcus gordonii* is known to inhibit microbial growth. However, *Aa* produces the enzyme catalase to detoxify H_2O_2, which is also utilized in the pathogen's respiration, increasing its fitness and virulence. *S. gordonii* also produces L-lactate, which *Aa* uses as a carbon source, creating a cross-feeding relationship that benefits the pathogen. Despite this knowledge, individuals harbor diverse communities of oral streptococci, thus variation in metabolite production across these different strains and species may influence interactions with *Aa*. Specifically, previous data shows that diverse streptococcal members have varying levels of fitness when cocultured with Aa as well as induce differential fitness of the pathogen. This project aims to compare H_2O_2 and L-lactate production across diverse oral streptococcal species in order to better understand these differing interactions. To accomplish this, we are using an optimized reflectometric assay that allows for accurate measurement and comparison of metabolite production across streptococcal members. Streptococcal cultures were grown in chemically defined media, washed to remove residual metabolites, back-diluted to mid-log phase, and normalized by optical density before metabolite measurements are collected using a reflectometric meter. Standard curves generated from known concentrations of H_2O_2 and L-lactate were then used to quantify metabolite concentration across species. Current results show differences in metabolite concentrations across diverse species and we expect that further work may help explain previously observed differences in their interactions with *Aa*. Understanding these metabolic relationships will provide insight into how variation across the oral streptococci influences the persistence of *Aa* and the microbial interactions that contribute to periodontal disease.

Introduction

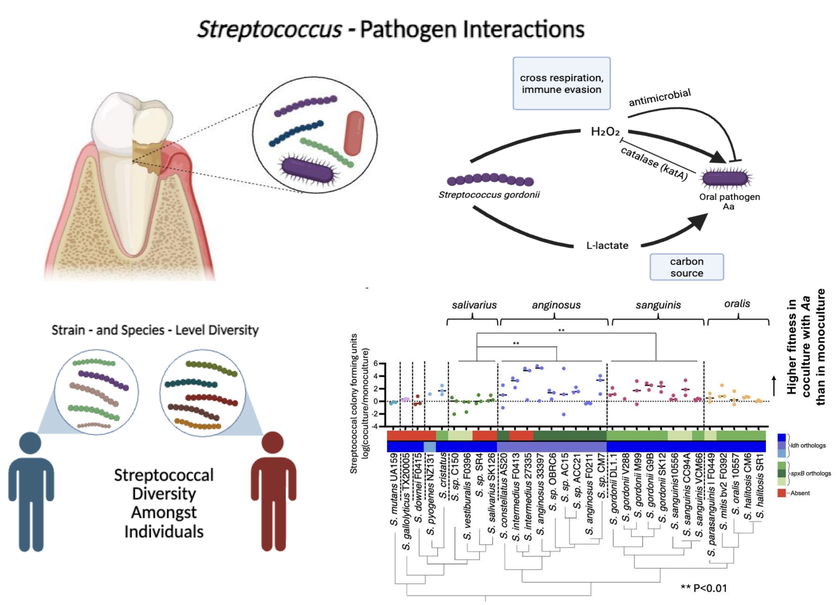


Fig. 1. Rationale for measuring metabolite concentration Oral streptococcal strains exhibit differential fitness during coculture with *Aa* compared with monoculture. Variation in *spxB* (hydrogen perodxide) and *ldh* (lactate) orthologs identified by pangenome analysis suggests that strain-specific metabolite production may underlie these fitness differences. Statistical significance was assessed using a Kruskal–Wallis test with Dunn's multiple-comparisons test. Figure made by Grace Heine.

Research Question: Are differences in H_2O_2 and L-lactate production in streptococci associated with strain-specific fitness differences during coculture with *Aa*?

Methods

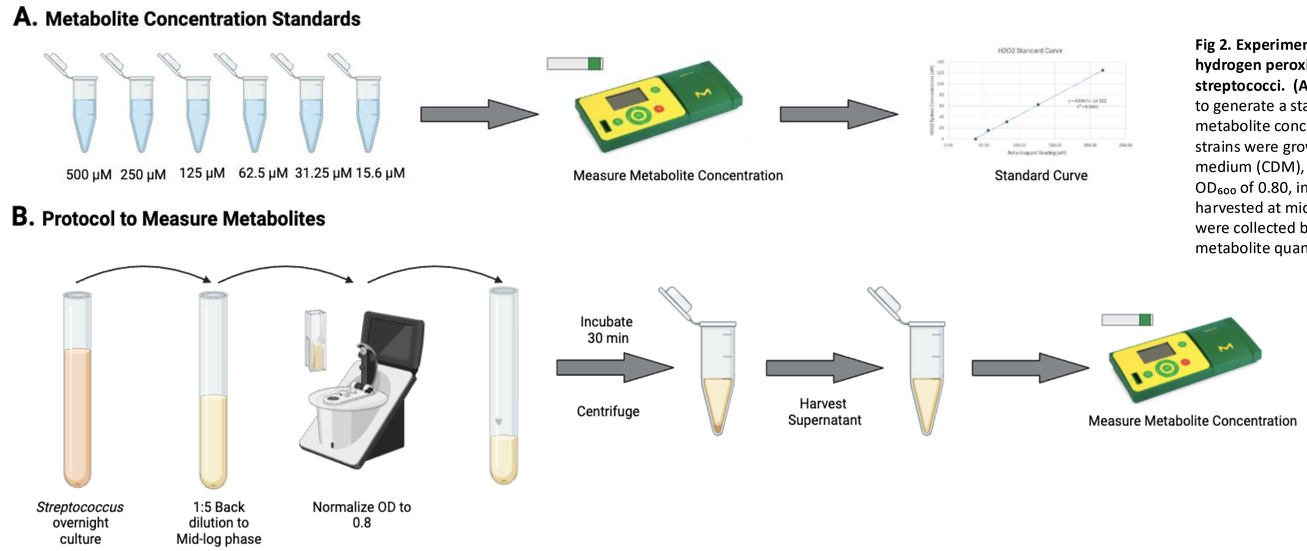


Fig 2. Experimental workflow for quantifying hydrogen peroxide and lactate production by oral streptococci. (A) Serially diluted standards were used to generate a standard curve with an $R^2 > 0.99$ for metabolite concentration. **(B)** Oral streptococcal strains were grown in liquid chemically defined medium (CDM), back-diluted 1:5, normalized to an OD_{600} of 0.80, incubated for 30 minutes and harvested at mid-log phase. Culture supernatants were collected by centrifugation and used for metabolite quantification using a reflectometer.

Results

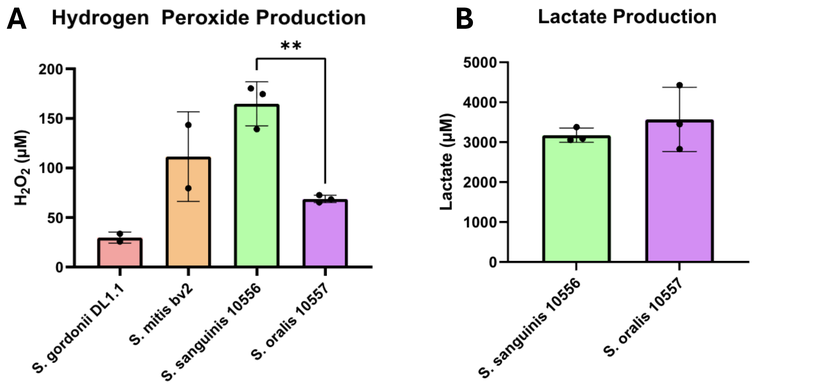
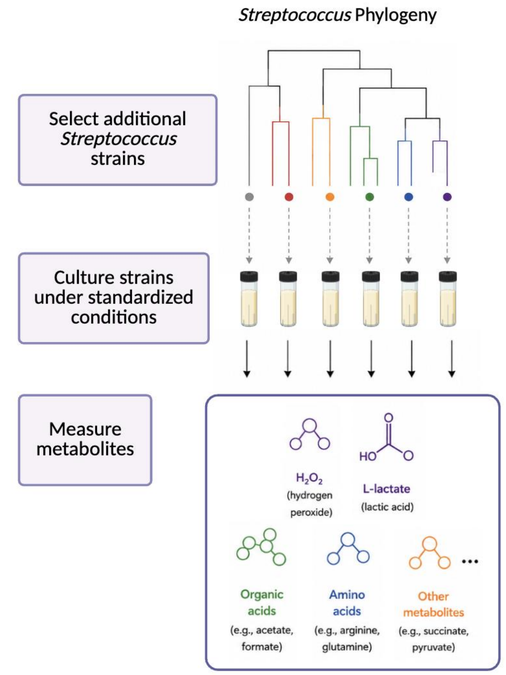


Fig 3. Metabolite production by oral streptococcal strains. Oral streptococcal strains were grown in liquid CDM to mid-log phase, normalized to an OD_{600} of 0.80, and incubated in fresh CDM for 30 minutes. Metabolite concentrations were quantified from culture supernatants. Bars represent the mean ± SD of the biological replicates. Each biological replicate represents the average of technical replicates. **(A)** Hydrogen peroxide production across four oral streptococcal strains. Statistical significance was assessed using ANOVA followed by Tukey's multiple-comparisons test, **$p < 0.01$. **(B)** Lactate production across oral streptococcal strains.

Conclusions

- **Oral streptococcal strains exhibited substantial differences in hydrogen peroxide production, with approximately a 6-fold range observed across strains.**
- **Lactate production was detected in both *S. sanguinis* and *S. oralis*, with comparable concentrations under the conditions tested.**
- **These findings suggest that metabolic output varies among oral streptococci and may contribute to strain-specific ecological interactions.**
- **Future studies will determine how these metabolic differences influence interactions with *Aa*.**

Future Directions



Continue screening additional oral streptococcal strains for H_2O_2 and L-lactate production

Expand metabolite profiling to identify metabolic phenotypes associated with previously observed differences in coculture fitness with *Aa*

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